

"Design by Experiencing": Supporting Early-Stage Driving Design Reasoning and Empathy Construction through Low-Cost Embodied Simulation

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Abstract

Early-stage design for driving-related systems is challenging because designers have limited access to real driving contexts, while common representations such as sketches and static prototypes fail to capture key experiential constraints, including motion, spatial perspective, and attentional demand. This paper examines how simplified embodied simulation can support early-stage design reasoning and empathy without relying on high-fidelity simulators. We present KarKit, a low-cost, modular toolkit that selectively materializes essential driving conditions such as physical space, movement continuity, environmental constraints, and basic embodied control. Through a design workshop with open-ended driving scenarios, we show that simplified embodied simulation supports situated understanding, shifts attention from interface details to experiential constraints, and enables empathy through embodied engagement. We discuss implications for embodied simulation toolkits and position simplified embodied simulation as an early-stage method within a staged design workflow.

CCS Concepts

• **Human-centered computing** → **Heuristic evaluations**; *Empirical studies in HCI*; User interface toolkits.

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Keywords

Embodied interaction, Early-stage design, Automotive interface design, Design methods, Empathy construction, Design workshops

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1 Introduction

Designing interaction concepts for driving-related systems in early stages is challenging because designers often lack first-hand access to real driving contexts, making it difficult to judge user needs and design feasibility [6]. Driving is a highly embodied and dynamic activity involving spatial perception, continuous motion, time-pressured decision-making, and shifting attentional demands [40], which requires designers to understand not only interface elements but also how drivers physically engage with situated contexts over time [10, 37]. Taken together, space, time, attention, and control execution together constitute the foundational structure of embodied driving experience [23]. In practice, early design commonly relies on sketches, storyboards, or static interface prototypes, which support ideation but often fail to convey key experiential constraints such as temporal progression, spatial perspective, and attentional demand [24], limiting designers' ability to anticipate real-world interaction and develop empathy for drivers [27, 32]. While driving simulators and mixed reality systems can approximate real driving in controlled settings [11, 28], high-fidelity setups depend



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on specialized hardware and complex configuration, making them unsuitable for rapid iteration in early design stages [29, 33].

This work investigates low-cost embodied simulation as an alternative for early-stage driving design reasoning and empathy construction. Rather than reproducing full driving fidelity, we focus on selectively materializing key experiential constraints related to time, space, and attention. We introduce **KarKit**, a modular, low-cost embodied simulation toolkit, and examine its use through a design workshop. Our contributions are threefold: we articulate simplified embodied simulation as a method for early-stage driving-related design and clarify its role relative to high-fidelity simulation; we identify recurring shifts in designers' reasoning when engaging with simplified embodied simulation, highlighting how temporal, spatial, and attentional constraints become salient; and we derive design implications for embodied simulation toolkits, including selective materialization, mid-fidelity manipulability, and support for scenario-based design discourse.

2 Related Work

2.1 Driving Simulation in Automotive Interaction Research

Driving simulators are widely used in automotive HCI to study driving behavior, cognitive workload, and in-vehicle interaction [15, 25]. While high-fidelity systems provide strong support for evaluation and safety research, recent work highlights trade-offs among realism, feasibility, and participant experience [35, 42]. Their reliance on specialized hardware, complex setup, and predefined scenarios limits rapid iteration and exploratory ideation in early design stages [31, 36], leaving a gap between abstract early prototypes and high-fidelity simulation.

2.2 Early Prototyping for Embodied Interaction

Early-stage prototypes are primarily used to explore design directions and clarify assumptions rather than validate final systems [3]. Common representations such as sketches, storyboards, and screen-based prototypes support ideation but provide limited access to embodied and situated experience [39]. When interaction involves continuous movement, changing spatial perspectives, or attentional conflicts, these representations often fail to convey key experiential constraints [20], making it difficult for designers to anticipate real use and develop empathy in dynamic contexts such as driving [9, 26]. Prior work has explored experiential prototyping and mixed reality approaches [16], as well as design toolkits that externalize key experiential elements to support design reasoning and reflection [7, 19, 41]. Accessible and modular toolkits, in particular, have been shown to support early-stage exploration and shift attention from interface details to higher-level experiential structures [21], positioning them as bridges between abstract representations and situated experience.

3 KarKit: System Design and Development

KarKit was informed by a formative user study with designers, including questionnaires and semi-structured interviews, examining early-stage prototyping and empathy building in complex contexts

such as driving. The study showed that existing tools provide limited support for embodied understanding: high-fidelity simulators are costly and inflexible for rapid exploration, while text-, image-, or video-based descriptions lack actionability and bodily engagement. Designers reported that early-stage work benefits less from precise simulation than from experiences that preserve key embodied and dynamic conditions while remaining easy to construct and adjust.

Based on these findings, KarKit was designed to support *simplified embodied simulation* through low-cost, reconfigurable components. Rather than pursuing full-fidelity vehicle or environment modeling, the design goal was to instantiate only key experiential constraints relevant to early-stage reasoning. This design orientation is grounded in a selective materialization framework, in which motion continuity, spatial layout, and attentional demand are translated into manipulable system elements.

3.1 System Architecture

To operationalize this design goal, KarKit adopts a modular hardware–software architecture. Rather than tightly integrating components into a fixed system, the design emphasizes constructability and reconfigurability of situational conditions by separating an *interaction space* for embodied control and perception from a *test space* where a small-scale vehicle operates within a configurable physical environment. Embodied input, vehicle motion, and first-person visual feedback form a continuous action–feedback loop across these spaces.

At the hardware level, the system consists of a custom-built vehicle unit driven by a microcontroller and motor driver shields, a wireless communication module, and a camera for first-person visual feedback. The physical test space is constructed using modular environmental materials that can be rearranged to represent different road and situational conditions.

At the software level, KarKit adopts a layered control structure consisting of a vehicle control layer (Arduino-based), a wireless communication layer (ESP32-based), and a web-based client interface. Control commands are transmitted from the client interface to the vehicle controller via wireless communication and serial protocols. The custom-built vehicle provides open interfaces and adjustable motion parameters, supporting modular modification and experimental controllability beyond off-the-shelf platforms.

3.2 Function Modules

KarKit consists of four interrelated modules defined by the situational conditions they materialize within simplified embodied simulation: the **basic driving simulation module**, which provides continuous motion and first-person visual feedback through a camera-equipped vehicle and lightweight control application; the **cockpit simulation module**, which supports embodied operation and attentional engagement via steering, acceleration, and braking controls; the **environment simulation module**, which enables rapid construction and comparison of environmental conditions such as pedestrians, terrain, lighting, and signage; and the **vehicle configuration module**, which allows designers to explore how changes in vehicle state affect driving experience while keeping the overall system structure stable.

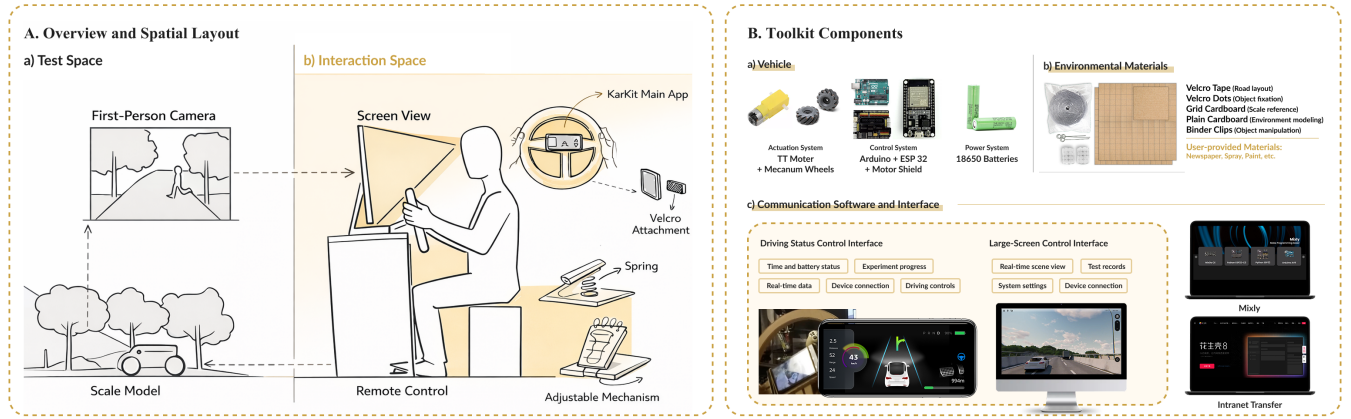


Figure 1: KarKit overview and toolkit components. (A) Overview and spatial layout. The system consists of a *Test Space* for physical vehicle movement and an *Interaction Space* for embodied control and perception. (B) Toolkit components. The toolkit includes a vehicle unit, environmental materials for scenario construction, and communication software and interfaces for control, monitoring, and configuration.

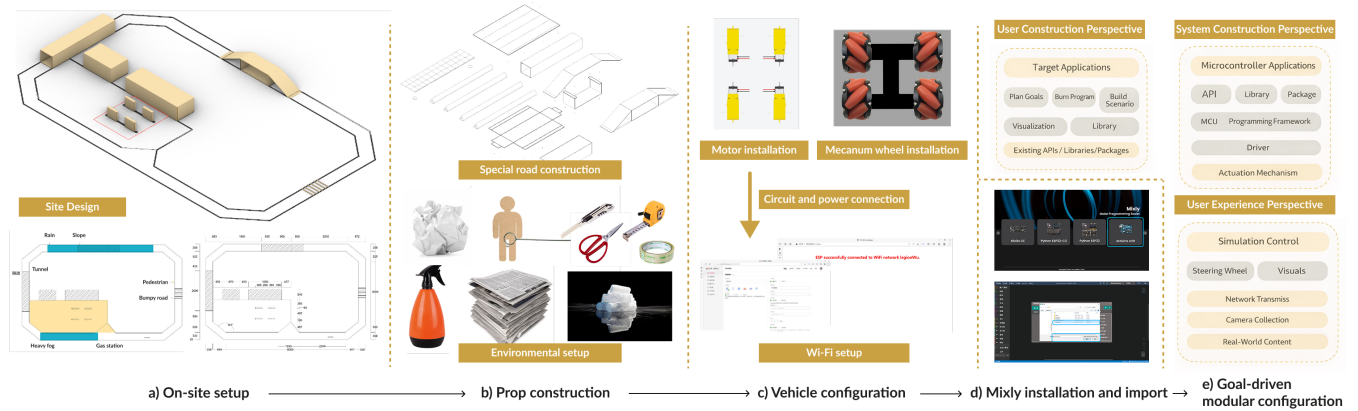


Figure 2: KarKit usage flow. The workflow spans situational setup, system connection, experience enactment, and iterative adjustment, supporting rapid configuration and reuse across design questions.

3.3 Usage Workflow

KarKit supports rapid deployment and repeated use through a flexible four-stage workflow: **situational setup**, in which designers construct simplified driving scenarios using low-cost physical materials; **system connection**, where the vehicle is placed into the scenario and connected via the KarKit application to enable control and first-person feedback; **experience enactment**, during which designers engage in embodied driving using cockpit controls; and **iterative adjustment**, allowing scenarios, environmental elements, and vehicle configurations to be modified and re-experienced in multiple cycles for comparison and early-stage exploration.

4 User Study

To examine how simplified embodied simulation is used in practice, we conducted a design workshop study with KarKit. The study

focused on how designers engaged with the toolkit during early-stage exploration, and how embodied experience shaped their understanding of driving situations and design problems.

4.1 Participants

We recruited $N = 12$ participants, including graduate students, senior undergraduates, and early-career practitioners with backgrounds in interaction or product design. All participants had two to six years of design training or practice and were familiar with common early-stage methods such as sketching, storyboarding, and low-fidelity prototyping. All had everyday experience as drivers or passengers. Some had prior exposure to transportation- or cockpit-related topics through coursework or projects, but none had formal training with driving simulators. Driving proficiency was not used as a selection criterion.

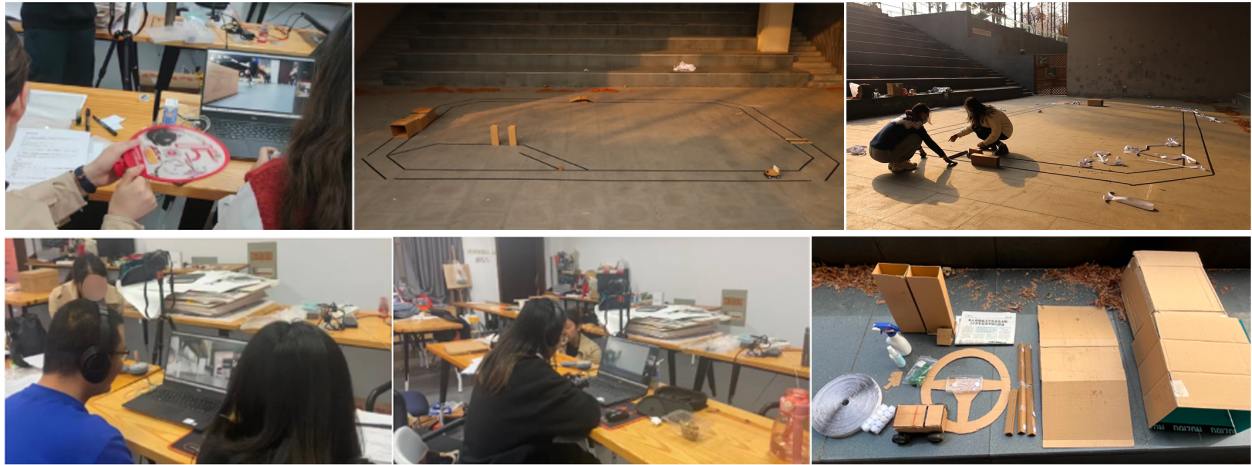


Figure 3: Design workshop. Key moments include physical scenario construction, embodied interaction with the system, collaborative discussion, and the use of low-cost materials to configure driving situations.

4.2 Procedure

The study was conducted as a 90-minute design workshop facilitated and documented by the research team. The workshop consisted of four stages: introduction and context setting, system familiarization, scenario-based design exploration, and reflective discussion.

Introduction and Familiarization. Researchers introduced the study goals and provided a brief overview of KarKit’s components and basic usage, avoiding detailed instruction. Participants then freely explored vehicle control, first-person visual feedback, and environment construction without a predefined task.

Scenario-Based Exploration. Participants worked in groups of four on open-ended driving-related scenarios (e.g., attentional distraction, environmental interference, atypical road conditions). Using KarKit, they constructed situations and explored design ideas through embodied interaction, scenario configuration, sketching, and discussion.

Reflection. The workshop concluded with a reflective discussion on participants’ design processes, changes in situational understanding, and the perceived value and limitations of simplified embodied simulation in early-stage design.

4.3 Scenario-Based Design Process

To clarify how KarKit influenced concrete design reasoning, we describe one representative scenario from the workshop. Although the scenarios were open-ended, each group followed a structured template including (1) a design objective, (2) environmental configuration across weather–terrain–human dimensions, (3) role assignment such as observer and facilitator, and (4) predefined recording methods including subjective evaluation and observational notes. This structure ensured that exploration remained grounded in concrete design tasks rather than purely free experimentation.

In one scenario, the design question concerned how multimodal stimuli (e.g., voice prompts, visual media, environmental disturbances) affect drivers’ attentional focus during driving. Initially,

discussion centered on interface-level solutions such as adjusting notification layout and visual hierarchy.

Using KarKit, participants constructed a driving environment including a bumpy road segment, a slope, and pedestrian elements. During embodied driving, additional stimuli such as audio interaction and environmental changes were introduced. Participants reported that attention shifts were strongly influenced by motion instability and spatial constraints, rather than by interface layout alone.

As a result, the design discussion shifted from visual arrangement to context-sensitive timing strategies. Participants proposed reducing information density during high-motion or visually constrained segments and deferring non-critical prompts under complex road conditions. This scenario illustrates a complete reasoning process in which embodied experience prompted a reframing of design constraints and a corresponding shift in design direction.

4.4 Analysis

We conducted a qualitative thematic analysis of workshop data, including video and audio recordings, participants’ verbal expressions, and researchers’ field notes. All data were transcribed and anonymized. Two researchers independently coded the data using an iterative process, focusing on embodied experience, design reasoning, and shifts in problem framing. Codes were compared and synthesized into higher-level themes, with disagreements resolved through discussion and reference to the original data.

5 Findings

5.1 Experiencing Driving as a Situated Activity Rather than a Game

Some participants initially framed KarKit as similar to driving games or screen-based simulations. This perception shifted once they experienced continuous vehicle movement in real physical space. Physical paths, turns, and spatial boundaries directed attention toward the driving situation itself rather than the system

representation. As one participant noted, “Once it’s really moving in space, I start thinking about what it would be like to drive here.” Continuous motion in physical space enabled designers to engage with driving as a situated activity rather than an abstract simulation.

5.2 Embodied Controls Make Attention and Cognitive Load Visible

Using the cockpit simulation module made designers more aware of how bodily actions shaped attention and cognitive load. Although the controls were simplified, physically performing driving actions revealed moments of distraction, overload, and attentional switching. This shifted design discussions from interface elements to operation rhythm, timing, and potential cognitive pressure.

5.3 Making Context and Empathy Experiential

The environment simulation module allowed designers to translate abstract situational assumptions into directly experienceable conditions. By modifying environmental elements such as pedestrians or lighting, participants could immediately sense how context affected driving behavior and decision-making. These experiences supported more context-sensitive reasoning and grounded discussions about drivers’ needs.

5.4 Treating Fidelity as a Design Variable

Participants treated KarKit’s fidelity as adjustable rather than fixed. Instead of seeking realism, designers tuned vehicle states and environmental conditions to match their current design questions. This configurability supported comparison across conditions and positioned KarKit as a tool for exploration rather than validation, helping designers generate early-stage design assumptions.

6 Discussion

6.1 Why Driving Matters as a Research Setting

Driving concentrates key constraints of embodied interaction, including continuous motion, time pressure, and attentional competition. Workshop results show that preserving real physical space, continuous motion, and basic operational feedback was sufficient for designers to reason about attention, visibility, and action pacing. This aligns with work framing embodied simulation as grounded in action and perception [18], and suggests that full-fidelity simulation is not required for experiential understanding in early design. Given the safety and ethical barriers to real-world driving access, simplified embodied simulation offers a practical entry point.

6.2 What Simplified Embodied Simulation Is Good for and Not for

Workshop findings indicate that simplified embodied simulation complements early-stage design by supporting experiential understanding that informs subsequent design work [8]. Its applicability depends less on domain and more on whether a situation is highly embodied and constrained by space and time. While immersive simulations can support empathy and design decisions [5, 13, 14, 22, 30, 34], simplified embodied simulation does not aim for precise reproduction. Instead, selectively simulating key experiential

conditions was sufficient for early-stage empathy and reasoning. Due to its simplified feedback, however, it is not suitable for final usability evaluation or behavioral prediction.

6.3 Design Implications for Building Embodied Simulation Toolkits

Our findings suggest several implications for toolkit design. First, toolkits should emphasize goal-oriented selective simulation rather than comprehensive replication, consistent with mixed-reality prototyping approaches [38]. Second, moderate fidelity combined with high manipulability supports exploratory comparison and adjustment [4]. Third, embodied simulation should support scenario-based design discourse, as empathy and experiential grounding are central to design thinking [2, 12, 17, 31]. Finally, modular structures should enable rapid parameter variation and comparison to support exploration in complex design spaces [1].

7 Limitations and Future Work

This study is based on a design workshop with a limited number of participants, and the findings primarily reflect designers’ experiential engagement and reasoning in early-stage exploration rather than quantitative design outcomes. The workshop was structured as a methodological exploration rather than a controlled comparison study. We did not include a baseline condition (e.g., sketching- or storyboard-only exploration), as the primary goal was to examine how simplified embodied simulation shapes design reasoning patterns rather than to measure performance gains relative to traditional methods. KarKit was intentionally designed as a simplified embodied simulation, and does not support precise behavioral prediction or final usability evaluation; therefore, the results do not generalize to later-stage validation or safety assessment. Accordingly, the study does not claim demonstrable improvement in design effectiveness, but instead provides qualitative insight into how embodied experience reframes early-stage design thinking.

In addition, while driving effectively foregrounds temporal, attentional, and bodily constraints, it represents only one class of embodied design contexts. Future work could examine simplified embodied simulation as an early step in a staged design pipeline, in combination with higher-fidelity simulation or real-world testing, and explore how design assumptions transfer across fidelity levels. Future research may also adopt comparative experimental designs to evaluate incremental value relative to established early-stage prototyping approaches. The approach may be extended to other safety-critical embodied domains, such as aviation, medical procedures, or industrial systems, to further clarify its applicability and boundaries.

8 Conclusion

This work examines simplified embodied simulation as an early-stage design method. Through the development of a low-cost, modular toolkit and a structured scenario-based workshop study, we show that selectively materializing key experiential constraints can reshape how designers frame and reason about situated interaction problems without requiring high-fidelity simulation. Using KarKit as a method instantiation, we identify recurring shifts in how designers perceive and reason about driving contexts. These findings

provide empirical grounding for simplified embodied simulation as a transferable approach for early design exploration and offer design implications for embodied simulation toolkits in complex, embodied contexts.

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